

Accounting for the spouse when measuring inequality of opportunity

Andreas Peichl¹ · Martin Ungerer²

Received: 27 May 2015 / Accepted: 23 July 2016 / Published online: 9 August 2016
© Springer-Verlag Berlin Heidelberg 2016

Abstract The existing literature on inequality of opportunity (IOp) has not addressed the question of how the circumstances and choices of spouses in a couple should be treated. By omitting information relevant to the spouse in IOp estimations, the implicit assumption has been full responsibility for the spouse's income, effort and circumstance variables. In this paper, we discuss whether or not the spouse's characteristics should be treated as responsibility factors. Using German micro data, we analyze empirically, how IOp estimates are affected when a spouse's circumstance or effort variables are included in the analysis. We find that including spousal variables can increase IOp measures by more than 20 (35) percent for gross (net) earnings. The less responsibility assumed for the partner's variables, the higher the IOp estimate.

Andreas Peichl and Martin Ungerer are grateful to two anonymous referees, Marc Fleurbaey (the editor), Rolf Aaberge, Paul Hufe, Vito Peragine, Dirk Neumann, Nico Pestel, Sydni Pierce, John Roemer, Arna Vardardottir and seminar and conference participants at the Canazei Winter School, ECINEQ Luxembourg, in Mannheim and London for many valuable comments and suggestions.

Electronic supplementary material The online version of this article (doi:[10.1007/s00355-016-0985-9](https://doi.org/10.1007/s00355-016-0985-9)) contains supplementary material, which is available to authorized users.

✉ Andreas Peichl
peichl@zew.de
Martin Ungerer
ungerer@zew.de

¹ ZEW, University of Mannheim, IZA and CESifo, L7, 1, 68161 Mannheim, Germany

² ZEW and University of Cologne, Cologne, Germany

1 Introduction

The concept of equality of opportunity (EOp) has received considerable attention since the seminal contributions of [Roemer \(1993, 1998\)](#), [Van de gaer \(1993\)](#) and [Fleurbaey \(1995\)](#).¹ The EOp literature is interested in the sources of inequality, distinguishing between exogenous circumstances and (partially) endogenous effort. Circumstances are defined as all factors beyond the sphere of individual control, such as parental background or gender, for which society deems that individuals should not be held responsible. Conversely, effort refers to all factors for which individuals are held responsible because they (partly) control or choose them, e.g. decisions concerning schooling or labor supply. Inequalities due to circumstances call for compensation whilst society considers inequalities due to effort to be legitimate. All previous studies of inequality of opportunity (IOp) have exclusively analyzed the impact of *individual* circumstances and choices. A factor which has not yet been studied in the literature is the relation between the personal characteristics of spouses in couples and IOp.² By omitting spousal information in IOp estimations, the implicit assumption in previous studies was full responsibility for the spouse's income, effort and circumstance variables. In this paper, we discuss whether or not the spouse's characteristics should be treated as responsibility factors and empirically investigate the impact on IOp estimates of explicitly incorporating the spouse's characteristics in the analysis.

Undertaking this exercise is no trivial task, at least from a philosophical point of view. The key question with relation to the EOp literature is, what should individuals be held responsible for? According to EOp theory, individuals are not responsible for their circumstances but only for their effort variables. The same is true for the spouse; he or she is not responsible for his or her circumstances but only for his or her choices. Choosing a partner, however, is a choice. Given that a spouse's circumstances are (usually) known when making the mating decision, one could argue that one should be held responsible for the circumstances of one's spouse. In fact, due to the omission of spousal variables from the analysis, this is the implicit assumption in the existing literature. However, this could be at odds with the basic EOp notion that individuals should not be held responsible for factors beyond the sphere of individual control. In other words, one could question whether it is acceptable to hold someone responsible for his or her spouse's circumstances when the spouse himself is not held accountable for their own circumstances. For effort variables, the decision becomes even more tricky. The key question here is how choices are made within couples: Do spouses make joint decisions (unitary household model) or is it rather the case that spouses bargain between their individual choices (collective household model)? In the former case, one could be held (more) responsible for the choices of one's spouse than in the latter case—especially when decisions change after marriage. We discuss these

¹ See, e.g., [Ramos and Van de gaer \(2016\)](#), [Roemer and Trannoy \(2016\)](#) or [Ferreira and Peragine \(2015\)](#) for recent surveys.

² In contrast, studies on inequality within couples look at inequality of outcomes rather than IOp. For instance, [Lise and Seitz \(2011\)](#) show that standard measures of inequality in terms of consumption are underestimated by about 50 percent, if one neglects intra-household inequality. For a survey, see e.g., [Browning et al. \(2013\)](#).

philosophical questions in further detail below and present four alternative scenarios demonstrating how a spouse's characteristics might be handled. Choosing among these scenarios entails a distinct normative choice. We empirically analyze whether, and to what extent, this choice matters for IOp estimates.

As mentioned above, by looking only at one's own circumstances, the implicit assumption in the literature produced thus far has been full responsibility for spousal variables independent of whether they constitute circumstance or effort factors. In the empirical part of this paper, we use German micro data in order to investigate the extent to which relaxing this assumption will result in changes in IOp estimates. Our results show that including spousal variables increases IOp measures by more than 20 (35) percent for gross (net) earnings. The less responsibility assumed for the partner's variables, the higher the IOp estimate. Our findings therefore suggest that deciding whether and how to account for spousal outcomes in couples is of considerable importance for IOp estimates. Assuming full responsibility, the current practice in the literature might thus result in underestimation of IOp.³ We show that at least part of this effect originates from assortative mating, suggesting that sorting into couples cannot be ignored when measuring IOp.

This paper is organized as follows: In Sect. 2, we introduce the conceptual framework for measuring IOp with couples. Section 3 describes the data. Section 4 presents the results of our empirical analysis. Section 5 concludes.

2 Conceptual framework and methodology

2.1 Measuring IOp: a simple model

In order to compare our results to previous IOp estimates, we follow standard practice to define our theoretical and empirical approaches.⁴ In accordance with [Roemer \(1998\)](#), we distinguish between two generic determinants of outcome y_i of individual i . First, circumstances C_i are characteristics which are beyond the scope of individual control (consider race, gender, family background) and which are therefore a source of illegitimate inequalities in outcomes. Second, effort E_i represents all factors affecting earnings which are assumed to be the result of personal responsibility.⁵ We focus on annual labor earnings w_i which is generated by some function f of C_i and E_i , which itself might depend on C_i : $E_i = E_i(C_i)$.

$$w_i = f(C_i, E_i(C_i)). \quad (1)$$

³ This is even more problematic when estimating IOp for societies where marriages are arranged (e.g., by parents) and where partners themselves have a limited say in who they marry.

⁴ The notation closely follows [Niehues and Peichl \(2014\)](#).

⁵ As is common in the majority of EOp literature, we do not explicitly take into account the role of luck in our estimations. Hence, we (implicitly) assume that luck belongs to the sphere of individual responsibility and in our deterministic model, the individual is held responsible for any random component that may affect his income and that cannot be attributed to the observed circumstances. The same is true for potential measurement errors in the earnings data. See [Lefranc et al. \(2009\)](#) for the extension of the EOp framework in order that it explicitly take luck into account.

Following [Ferreira and Gignoux \(2011\)](#) and [Niehues and Peichl \(2014\)](#), we employ the ex-ante approach of EOp⁶ by partitioning the population of agents $i \in \{1, \dots, N\}$ into a set of disjunct types $\Pi = \{T_1, T_2, \dots, T_k\}$, i.e. subgroups of the population which are homogeneous in terms of their circumstances. The income distribution within a given type is a representation of the opportunity set which can be achieved for individuals with the same circumstances C_i by exerting different degrees of effort. Following the utilitarian reward principle ([Fleurbaey and Peragine 2013](#); [Ramos and Van de gaer 2016](#)), perfect EOp is achieved if the mean advantage levels μ are identical across these types, i.e., $\mu^k(w) = \mu^l(w)$, $\forall l, k | T_k, T_l \in \Pi$. In our case, this corresponds to identical mean wages across types. Measuring IOp thus means capturing the extent to which $\mu^k(w) \neq \mu^l(w)$, for $k \neq l$. To compute a measure of IOp, [Checchi and Peragine \(2010\)](#) suggest constructing a hypothetical smoothed distribution ([Foster and Shneyerov 2000](#)), $\mu^k(w)$, which is obtained when each individual outcome, w_i^k , is replaced by the group-specific mean for each type, $\mu^k(w)$.

Based on this smoothed distribution, we calculate for any (scale invariant) inequality index I the absolute IOp level (IOL) $\theta_a = I(\{\mu_i^k\})$. While the idea of opportunity egalitarianism is to look at IOp and not at total inequality in outcomes, it might still be informative to look at the relative share of total inequality that can be attributed to circumstances, i.e. the IOp ratio (IOR) $\theta_r = \frac{I(\{\mu_i^k\})}{I(w)}$. This approach allows the total income inequality to be decomposed into inequality within types (i.e. effort inequality) and inequality between types (i.e. opportunity inequality). Due to data limitations, we restrict our analysis to between type inequality since cell sizes of types tend to become smaller with increasing numbers of regressors. In order to respect the axioms of anonymity, Pigou-Dalton transfer principle, normalization, population replication, scale invariance and subgroup decomposability, we choose a member of the Generalized Entropy class ([Shorrocks 1980](#)) as our inequality measure (see, e.g., [Ferreira and Gignoux \(2011\)](#) and [Ramos and Van de gaer \(2016\)](#) for a discussion of measures and axioms). By introducing the further requirement of path-independent decomposability (see [Foster and Shneyerov \(2000\)](#) for a thorough discussion of this and the other axioms the GE measures satisfy), the set of eligible indices reduces to the mean log deviation (MLD) $I_0 = \frac{1}{N} \sum_i \ln \frac{\mu_w}{w_i}$.

2.2 Responsibility for the mating choice?

The key question in the literature on EOp is what individuals should be held responsible for, i.e. where to draw the responsibility cut-off (see, e.g., the discussion in [Ramos and Van de gaer \(2016\)](#), [Roemer and Trannoy \(2016\)](#) or [Ferreira and Peragine \(2015\)](#)). At the extremes of the range of potential cut-offs, hard determinists, on one end, would deny the existence of free will and the resulting set of responsibility factors

⁶ In the (empirical) EOp literature, two different approaches have been used to estimate IOp ([Fleurbaey and Peragine 2013](#)): ex-ante vs. ex-post. The former partitions the population into types, i.e. groups of individuals endowed with the same set of circumstances, and IOp is measured as inequality between types. In the latter case, individuals are classified into responsibility groups (tranches) of individuals at the same effort level and inequality within tranches is investigated.

would be empty. On the other end, following the self-ownership argument by Nozick (1974), individuals are entitled to the products of all personal characteristics (including genetic characteristics such as innate talent) and the set of circumstance factors would be empty, i.e. all inequalities based on legal activities are legitimate. Between these two extremes, two main concepts of responsibility emerge in the EOp literature (Ramos and Van de gaer 2016). On one hand, individuals could be held responsible only for what lies within their control (Arneson 1989; Cohen 1989; Roemer 1993, 1998). On the other hand, individuals could be held responsible for their preferences and the choices that follow from those preferences (Dworkin 1981a, b; Parijs 1995; Fleurbaey and Maniquet 2008).

As mentioned above, we follow Roemer (1998) and assume that individuals should not be held responsible for circumstances beyond the sphere of individual control. An open question following from this assumption in the EOp literature is, however, whether an individual in a relationship is responsible for his or her partner. So far, the literature does not explicitly consider the impact of partners, i.e. in empirical applications only *own* circumstances are included. Therefore, the current baseline case is full responsibility for the spouse's effort and circumstance variables [henceforth labeled as (i) *Full responsibility*] as these variables are implicitly treated as (unobserved) effort. We use this case as a benchmark in order to compare various possible scenarios which differ in terms of responsibility for the spouse's income, effort and circumstance variables. This baseline case can be rationalized as follows. Individuals are not responsible for their own circumstances but only for their own effort variables. The same is true for the spouse. He or she is not responsible for his or her circumstances but only for his or her choices. Choosing a partner, however, is a choice. A spouse's circumstances are (usually) known when committing to a relationship and are therefore an individual choice in the sense that individuals are fully aware of the pre-determined circumstances of their partner. One can also argue that an individual is responsible not only for the spouse's circumstances, but also for the spouse's choices. This is the case in an unitary household model where the household acts as one unit and both partners jointly decide on, e.g., labor supply, effort, and hence income (Chiappori and Meghir 2015).

It is true, at least in developed Western societies, that the "choice" of a partner is indeed considered to be a true individual *choice*. However, from a historical perspective this is a relatively recent development. Furthermore, while the mating itself can be argued to be a choice, the interpretation for the effort variables of the partner is less clear as these are subject to potential changes after marriage. Hence, they are arguably outside the sphere of individual control of the spouse. In addition, the circumstances and effort of the partner may affect the individual's own effort and may therefore have an indirect effect on own earnings. On the one hand, well educated or hard working spouses may motivate their partner to engage in similar activities, while low effort partners may discourage their spouses from exerting higher degrees of effort. On the other hand, individuals with hard working partners may feed from their spouse's effort by adjusting their labor supply. In the following paragraphs, we present three relaxations of the baseline case assumption of full responsibility. In this respect, we analyze to what extent accounting for partners' circumstance and effort variables matters when measuring IOp.

The first deviation from the baseline case can be rationalized as follows. It is well documented that wage setting institutions and potential wage discrimination are important determinants of individual earnings. While these factors might (or might not) be known to the individual, there is little he or she can do to influence them. Moreover, spouses' earnings are correlated due to assortative mating (Schwartz and Mare 2005; Pestel 2016) and, at least in Germany (but also France and the US), directly related due to joint taxation. The higher income of the first earner directly incentivizes the second earner to adjust his/her labor supply and income and vice versa. Hence, one can argue that individuals should not be held responsible for the spouse's wage earnings. Given, however, that circumstances are known, one could be held responsible for these factors. In fact, within a unitarian household framework, consumption of both spouses enters the joint household utility function in the same uniform way with identical welfare weights. The household is therefore assumed to maximize a common utility, leaving no room for bargaining or negotiations between household members. In such a framework, it can be argued that the members of an household acting as one unit are responsible for the joint actions and effort. We therefore define the second scenario as responsibility for the spouse's circumstances and choices, but not the spouse's wage [(ii) *Responsible for partner's circumstances and effort (unitary model)*].

While the unitary model of joint decision on labor supply and effort within a household is one possibility, the second scenario relies on the collective household model. Here, however, the consumption of each household member enters the utility function with potentially differing welfare weights, leaving room for differing preferences and intra-household bargaining. In this case, each partners individually exerts individual effort in terms of their labor market activity before bargaining about the distribution of joint household income. In such a framework, the effort variables of the partner are not the choice of the individual; rather, they are determined solely by the spouse, possibly after committing to a relationship. The third scenario is therefore responsibility for the partner's circumstances, but not for his or her income and choices (effort) [(iii) *Responsible only for partner's circumstances (collective model)*].

Finally, one can argue that individuals should not be held responsible for the circumstances, choices or income of their partner. There are two arguments which might support this scenario. First, the notion of IOp is that nobody should be held responsible for things beyond the sphere of individual control. As we cannot control the circumstances of our partner, except by leaving him or her or by refusing to mate, a partner's circumstances can thus not be considered as an own choice.⁷ Second, the choice set of available partners is not necessarily identical for all individuals. One observation of the existing literature on couples and family economics is the phenomenon of increased assortative mating. Hence, higher educated individuals increasingly mate with other highly educated individuals (Schwartz and Mare 2005). While high-skilled individuals may choose from a pool of both low- and high-skilled partners, potential partners of low-skilled individuals are mostly limited to other low skilled individuals. This leaves high and low skilled individuals with different choice sets. Combining both arguments,

⁷ In addition, one might not be fully aware of a spouse's full set of circumstances due to asymmetric information when committing to a relationship.

our fourth scenario is no responsibility whatsoever for the spouse's characteristics at all [(iv) *No responsibility*].

To sum up, the choice between these different levels of responsibility for one's partner cannot be answered without moral judgment and consideration of the context of the research question. In this paper we analyze whether these different scenarios differ empirically.

2.3 Empirical strategy to estimate IOp

2.3.1 Baseline case

In our empirical estimation approach, we use the same data, sample selection and parametric specification as [Niehues and Peichl \(2014\)](#) to estimate IOp.⁸ Relying on a parametric approach allows us to estimate the impact of numerous circumstance variables even in the presence of small sample and cell sizes.⁹ Log-linearization of Eq. (1) and adding an error term yields the following empirical specifications:¹⁰

$$\ln w_i = \alpha C_i + \beta E_i + u_i, \quad (2)$$

$$E_i = \kappa C_i + v_i. \quad (3)$$

Equation (2) represents the direct effect of circumstances on income while Eq. (3) models the indirect effect of circumstances on income through effort. Since it is unlikely that we will observe all relevant circumstance and effort variables that shape individuals' outcomes, estimating this model will likely yield biased estimates. However, in order to compute IOp shares, it is not necessary to estimate the structural model and to derive causal relationships. By substituting the effort Eq. (3) into the earnings Eq. (2), we obtain the following reduced-form relationship:

$$\ln w_i = \underbrace{(\alpha + \beta\kappa)C_i}_{\psi} + \underbrace{\beta v_i}_{\eta_i} + u_i. \quad (4)$$

⁸ In empirical estimations of EOOp, it is impossible to observe all characteristics that constitute an individual's circumstances (e.g. innate talent or ability). Hence, existing estimates of IOp are only lower bound estimates of the true share of unfair inequalities due to circumstances ([Ferreira and Gignoux 2011](#)). Exceptions are [Bourguignon et al. \(2007\)](#) who simulate the magnitude of omitted variable bias to estimate bounds around the true effect of observed circumstances on income inequality and [Niehues and Peichl \(2014\)](#) who suggest an upper bound estimator.

⁹ In contrast, non-parametric methods avoid the arbitrary choice of a functional form on the relationship between outcome, circumstances and effort (e.g. [Lefranc et al. \(2008\)](#), [Ferreira and Gignoux \(2011\)](#) or [Aaberge et al. \(2011\)](#)). However, this approach has the drawback that considering more than one circumstance variable is difficult due to practical reasons in the presence of small cell sizes which is usually the case in survey data. Access to large-scale administrative panel data with information on circumstances (family background), which is not available in Germany, would allow to estimate IOp also non-parametrically.

¹⁰ We use the log of incomes since estimation in logs is common in labor economics as log-normal is typically a very good fit for (right-skewed) earnings data. Nevertheless, we have also estimated the models in levels (as robustness checks; not reported) and did not find systematic differences.

This equation can be estimated by OLS to derive the fraction of variance which is explained by circumstances. If all observed circumstances, C^K , are included in the Eq. (4), then the estimate $\hat{\psi}$ measures the overall effect of circumstances on labor earnings, combining both the direct and indirect effects. On this basis, we can construct a parametric estimate of the smoothed distribution:

$$\tilde{\mu}^{LB} = \exp[\hat{\psi}C_i^K + \sigma^2/2]. \quad (5)$$

As we replace earnings by predicted earnings (with σ^2 being the estimated residual variance in the earnings equation, see Blackburn (2007)), all individuals with the same circumstances necessarily have the same advantage levels. Thus, in the case of absolute EOp, i.e. no income differences due to (observed) circumstances C_i^K , all predicted earning levels would be identical. IOp can subsequently be measured as the inequality of these counterfactual earnings levels, where remaining income differences are due only to differences in circumstances.

2.3.2 Spouses and IOp

We analyze the impact of partners' personal characteristics on IOp by extending the baseline measure in three steps. The current approach in the literature (according to Eq. (4)) implicitly assumes full responsibility for the spouse's variables as these are not included as circumstances in the regression and are hence treated as unobserved effort. This baseline case serves as a benchmark for further specifications (ii)–(iv). The second case assumes responsibility for the spouse's circumstances and effort variables [(ii) *Responsible for partners' circumstances and effort (unitary model)*]. This is empirically implemented by adding the earnings of the partner, $\ln w_i^P$, to Eq. (4), treating them as circumstances of the individual:

$$\ln w_i = \psi C_i + \zeta \ln w_i^P + \eta_i. \quad (6)$$

The definition of the third scenario [(iii) *Responsible only for spouse's circumstances (collective model)*] proceeds analogously. In order to hold the individual responsible for the circumstance variables of the partner, one must add the earnings and effort (E_i^P) variables of the partner to the wage regression (4):

$$\ln w_i = \psi C_i + \zeta \ln w_i^P + \lambda E_i^P + \eta_i. \quad (7)$$

Finally, the case of full responsibility adds all income, effort and circumstance (C_i^P) variables of the partner to the wage equation [(iv) *No responsibility*]. In this case, the partner is fully accounted for in the wage regression. Hence, all of his or her personal characteristics are treated as circumstances of the individual:

$$\ln w_i = \psi C_i + \zeta \ln w_i^P + \lambda E_i^P + \phi C_i^P + \eta_i. \quad (8)$$

To sum up, while in Eq. (4) all partner variables are (implicitly) treated as individual effort, this changes in the following equations. In the end, in Eq. (8), all partner

variables are treated as own circumstances. Hence, comparing Eq. (4) with (6)–(8) gives the full range of not accounting for spousal characteristics to fully accounting for spousal characteristics in IOP estimations.

Since in the literature on IOP measurement, individual as well as household income definitions are used, we replace individual income w_i with household (equivalized) income $y_h = (w_i + w_i^P + \omega_h)/m$ (where m is the modified OECD equivalence scale and ω_h represents non-labor household income) in an extension.¹¹ Thereby, we can explore to what extent our results are driven by intra household inequality.

3 Data

For our estimations, we use the 2013 version of the German [Socio-Economic Panel \(2013\)](#). The SOEP is a representative panel survey of households and individuals in Germany.¹² For our analysis, we use information from 1991 until 2011, i.e. the the period following the German reunification.

In line with the previous literature and especially following [Niehues and Peichl \(2014\)](#), we define our units of analysis as individuals aged 25–55 with data on parental background. We exclude singles from our analysis since we are interested in the relation between partners' characteristics in couples. The dependent variables are the logs of gross and net real labor earnings, adjusted by consumer price indices. In the sensitivity analysis, we extend the income definition to net equivalized household income using the modified OECD equivalence scale. Inequality measures are based on the corresponding absolute levels of earnings.

As *circumstance variables*, we include gender, the year of birth of the individual, dummies for whether the individual was born in a foreign country or in East Germany, the degree of urbanization of the place where the individual was born, categorical variables indicating the occupation and education of the individual's father as well as the height of the individual. The occupation of the father is categorized into five groups. The omitted benchmark is blue-collar. In addition, farmer, white collar, free-lancer, highly qualified white collar worker, and civil servant form separate groups. The education of the father is categorized into four groups, with no degree serving as the benchmark case. Secondary education, intermediate education, and higher education are used to define the other groups. Summary statistics on mean annual earnings and the main circumstance variables are reported in Table 3 in the Appendix. The *effort variables* used in the different specifications are work experience, working hours, education, and industry. Work experience is measured in years; working hours is measured in in weekly hours. The education of the individual is measured in years.

¹¹ Note that for this specification we cannot estimate Eq. (6).

¹² A detailed overview of the SOEP is provided by [Haiken De-New and Frick \(2005\)](#) and [Wagner et al. \(2007\)](#). Issues concerning sampling and weighting methods or the imputation of information in case of item or unit non-response is well documented by the SOEP Service Group.

In order to capture industry effects, eleven categories of industry are formed.¹³ A.1 gives a detailed description of the variables used.

4 Empirical results

4.1 Estimation of earnings equations

We begin our analysis by regressing the log of earnings for each year on all available circumstance variables which are expected to have an impact on labor earnings (Eq. (4)). The results for the baseline scenario as well as the case of no responsibility in gross and net annual earnings are reported in Tables B.4–B.11 in the Appendix. Using the baseline case (i), we can identify the well-documented gender wage gap in gross and net wages. Although the gender pay gap is gradually declining over time, women continue to receive significantly lower wages in comparison to male colleagues. Being an immigrant or being born in Eastern Germany has a negative impact on wages. Individuals with highly educated fathers or parents working as civil servants have higher wages compared to individuals with blue-collar or self-employed parents. Being born in larger cities, as opposed to the countryside, is also associated with higher wages. The educational degree of the father has a strong but ambiguous effect.

Controlling for the spouse's income in Eq. (6) yields the case of responsibility for spouse's circumstance and effort variables (ii). The income of the partner has a significant negative effect on personal earnings. At first glance, this seems to contradict previous findings of increased assortative mating. However, our analysis is carried out in terms of annual earnings. Taking into account the fact that share of part-time workers is about 40 percent for females and only 5 percent for men, the observed negative correlation is less surprising. In addition, we find a positive correlation for hourly wages. Comparing gross and net figures, we find systematically higher coefficients for net earnings. This indicates a possible effect of the German system of joint income taxation.

Controlling for effort and income variables of the partner, we assume responsibility only for spouse's circumstances (iii), as implemented in Eq. (7). A higher level of education on the part of the partner is associated with higher earnings of the individual. The coefficient increases over time and is significant since 1994. In line with the literature on assortative mating, we find a positive correlation between spouses' education. The industry in which the partner works has no clear effect on individual earnings. Figures for gross and net earnings show similar results for this specification.

The specification of no responsibility for the partner (iv) is implemented by controlling for all available partner information in Eq. (8). The partner's circumstance variables display different patterns compared to the individual's own variables. A higher educational degree of the father has a negative effect from 1991 until 1999, with significant values only in 1998 and 1999. Past 1999, the degree of the father has a

¹³ The eleven groups are public administration and social security, which serves as the benchmark, fishery and agriculture, energy, chemicals and steel, engineering, manufacturing, construction, wholesale and trade, transport, financial industry, service, education, and health service.

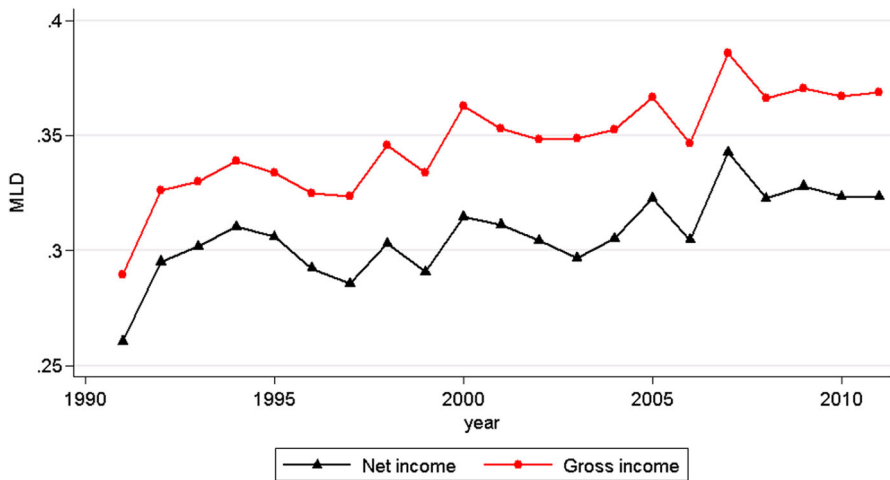


Fig. 1 MLD for annual income—gross vs. net

small and insignificant negative effect on earnings and has even turned positive since 2001. Having a foreign partner or a partner from East Germany has no significant effect on individual earnings. If the partner has a father working as a professional, e.g. as a highly skilled manager, this is associated with higher earnings, while a paternal occupation in agriculture corresponds to lower earnings.

4.2 Outcome inequality

Inequality in gross and net earnings as measured by mean log deviation (MLD) is depicted in Fig. 1 and shows a generally increasing trend for both gross and net earnings. Inequality increases from 0.28 to 0.36 between 1991 and 2011. The strongest increases are found between 1991/1992 and 2006/2007 with increases of 0.036 and 0.039 points, respectively. Inequality in gross earnings is significantly larger than in net earnings. Here, MLD increases from 0.26 in 1991 to 0.32 in 2011.

Comparing male and female gross earnings in Fig. 2, we find higher levels of inequality among female earners. However, while MLD for men is steadily increasing, it shows an inverse U-shaped pattern for women. The MLD for men increases from 0.12 in 1991 to 0.24 in 2011. Values for the female sample are 0.30 in 1991 and 0.33 in 2011, with a peak of 0.41 in 2004. Using net instead of gross earnings for the comparison of the male and female sub-samples, we find similar trends. The values in this case are, however, generally lower, as inequality in net earnings is lower than in gross earnings.

4.3 Inequality of opportunity ratios

The lower bound level of IOp (IOL) is estimated by computing the MLD of the predicted values from the earnings equation. We do this for specifications Eqs. (4) and

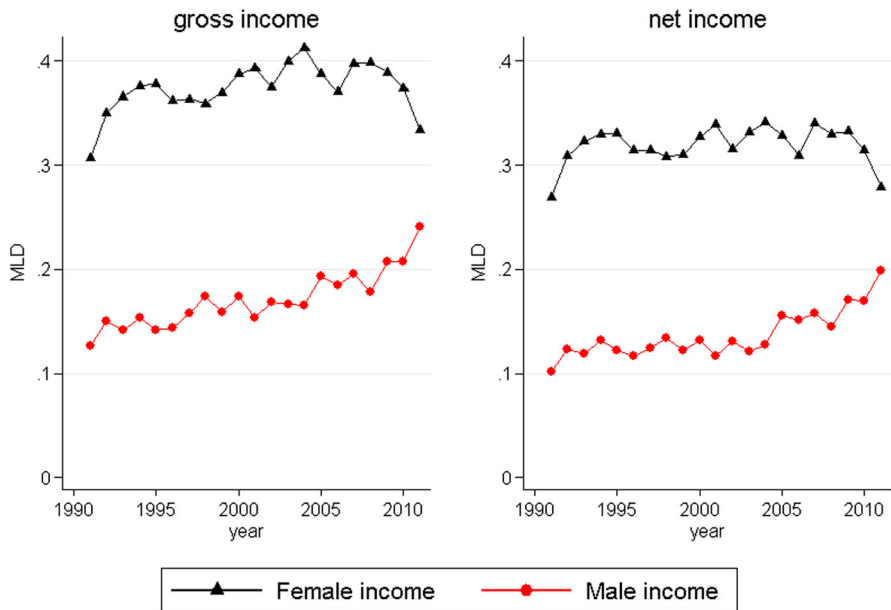


Fig. 2 MLD for annual income—male vs. female

(6)–(8), using gross and net annual earnings. The results are shown in Appendix A.3. Estimating the IOP ratio (IOR) is straightforward; we simply divide the IOL by total inequality in annual earnings. Starting with the comparison of gross and net annual earnings, we further distinguish between male and female sub-samples. The results for the full sample are shown in Fig. 3. The black line displays the baseline case from Eq. (4). The red line corresponds to the model which includes partner's circumstance variables shown by Eq. (6). The case of the model from Eq. (7), including both the spouse's circumstance and effort variables, is displayed by the blue line. Finally, the full specification including circumstances, effort and income variables of the partner corresponds to the yellow graph (Eq. (8)).

Looking at gross and net earnings displayed in Fig. 3, we find that the IOR in gross earnings was decreasing from 1991 until 2005. The majority of the decrease in the IOR can be said to have been driven by the increase in overall earnings inequality rather than a decreasing IOL. While the baseline case of full responsibility sees the IOR decrease from 45.3 to 37.5 percent, the specification using all available partner information sees the IOR decrease from 49.1 percent to 43.3 percent in 2003. Afterwards, we see an increase in IOR to 50.8 percent in 2008 and strong fluctuations thereafter until 2011. Our results show that accounting for spouse's variables can be important when measuring IOP, as including them increases the lower bound of IOP. While IOR for the baseline case is 38.1 percent in 2011, the full specification yields a value of 47.1 percent. Hence, IOR is 22 percent larger when all information available about the partner is taken into account. The less responsibility is assumed, the higher the IOP measure.

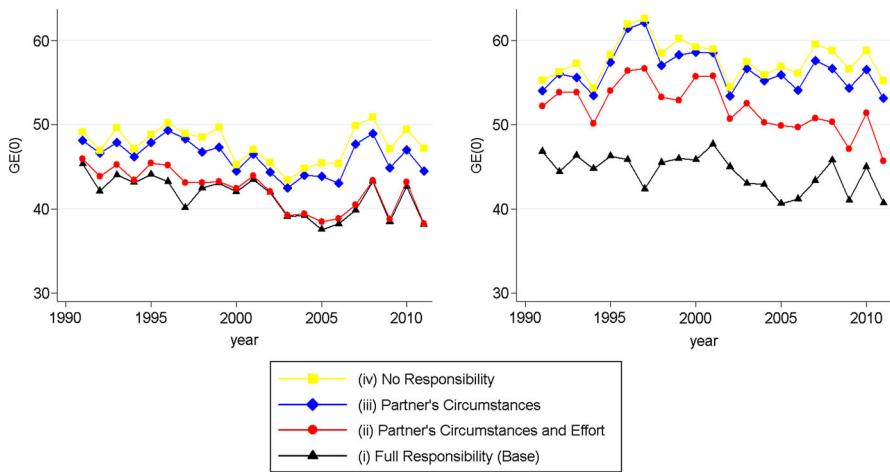


Fig. 3 IOR for annual income—gross vs. net

Looking at the baseline case for net earnings, we find a similar development, with slightly higher values. For 1991 and 2001, we find values of 46.8 and 47.7 percent, respectively. Following this, the IOR decreases until 2005 to 40.6 percent before increasing once again to 45.7 percent in 2008. After 2008, we can identify similar fluctuations in both net and gross earnings. When the effort and income variables of the partner are included, the IOR is fairly constant for net income and is higher than for gross income. Using the full specification, IOR has the same value of 55.2 percent in 1991 and in 2011, with an interim peak of 62.7 in 1997. The observation of a higher IOR in net earnings than in gross earnings can be explained by the lower overall inequality in net earnings, as the IOL remained on a similar level for both gross and net annual income. Our analysis indicates that including partner variables increases IOP by 8.3 (18.0) percent in 1991 and by 23.7 (35.6) percent in 2011 for gross (net) income. The general trend of a decreasing IOR in the baseline case is driven by the increase in earnings inequality rather than by a decrease in IOP.

The IOR for gross annual male earnings shows a generally u-shaped pattern over time. This can be seen in Fig. 4. The baseline case decreases from 36.2 in 1991 to 20.2 percent in 2005. There is a surprising peak of 30.7 % in 1999. Between 2008 and 2011, the IOR increased from 20 to 28 percent. We find a similar picture for the other specifications including the spouse's variables, but with higher values. The decrease under the full specification from 1991 to 1998, however, is less marked, declining from 41.9 to 35.8 percent. Only from 2000 to 2009, we find somewhat lower values between 31.1 and 32.9 percent. Thereafter, we once again see an increase to 43.5 percent in 2011.

For the female sub-sample, we find significantly lower initial values for the IOR in 1991. The baseline case yields 7 and the full specification 14 percent. The IOR increases over time, regardless of the specification used. As was the case for male earnings, there is a drop in the IOR between 1999 and 2000. Following this, however,

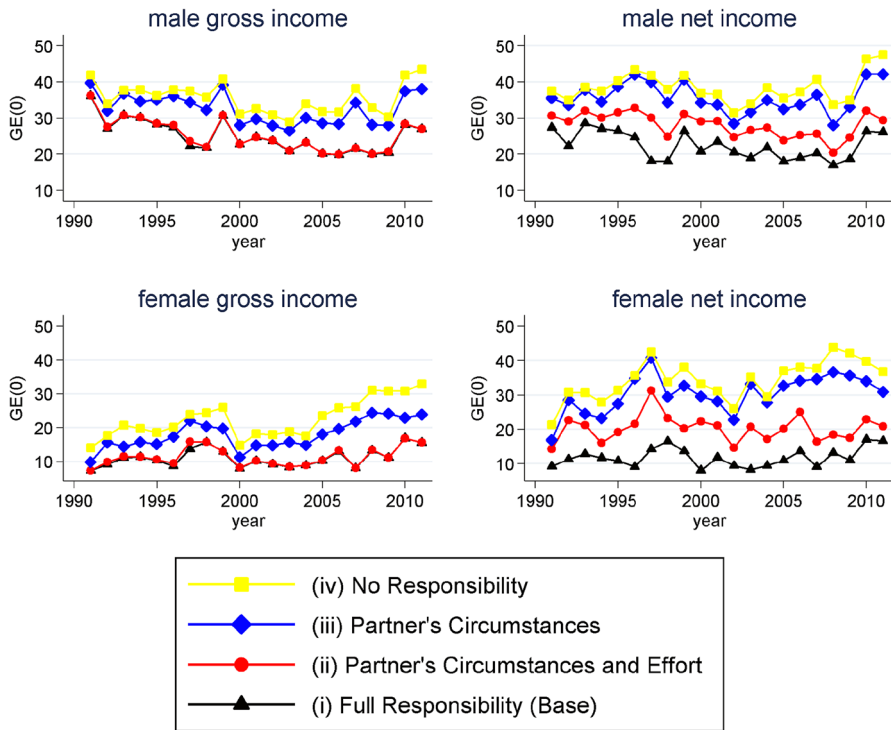


Fig. 4 IOR for annual income—male vs. female

the IOR once again increases to 15.6 in 2011 for the baseline case and 32.9 percent for the case of no responsibility.

Comparing male and female IOR for net earnings, the findings are similar to the case of gross earnings, with slightly lower figures. The only difference is in the case of no responsibility. While for men, the income of the spouse yields only limited additional information, IOR for female net earnings increases significantly when the income of the partner is included. IOR is 21.4 in 1991 and increases to 42.5 percent in 1997. IOR decreases to 26.0 in 2002, has a peak in 2008 and then once again decreases again to 36.8 percent in 2011. Adding partner variables to the analysis of gross income increases IOR for men by 15.9 (61.1) percent in 1991 (2011) and for women by 91.2 (111) percent in 1991 (2011).

To sum up, it is clear from our findings that the personal characteristics of the partner play a significant role for IOR. They increase our estimate of the IOR by up to 20 (35) percent for gross (net) earnings. Before analyzing the role of assortative mating in these results in Sect. 4.5, we check the sensitivity of our results with respect to several choices made in the next section

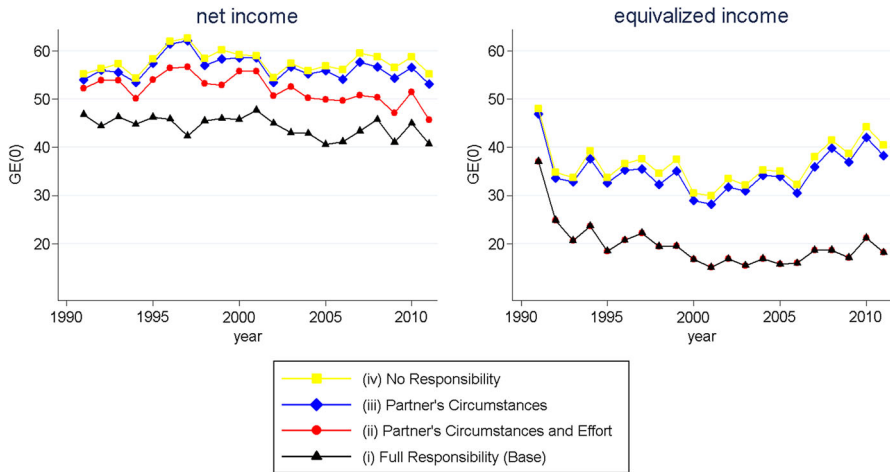


Fig. 5 IOR in $GE(0)$ for net equivalized income

4.4 Sensitivity analysis

Up to this point, our analysis has been limited to individual income using the MLD as our measure of inequality. In this section, we aim at overcoming these limitations by broadening our calculations to equivalized household income and applying several other inequality measures apart from the MLD.

In the literature on IOP measurement, individual as well as household income definitions are used. An analysis on an individual level focuses on the potential relationship between the circumstances and effort of the partner and individual outcomes. We do not explicitly control for joint decisions but rather implicitly take them into account by defining scenarios with different degrees of responsibility for the personal characteristics of the partner. It is, however, interesting to follow studies such as [Lefranc et al. \(2008\)](#) and [Ferreira and Gignoux \(2011\)](#) by using equivalized household income as the dependent variable. In this way, we can explore to what extent our results are driven by intra household inequality. However, in order to carry out our analysis using equivalized income, we have to alter the previously defined scenarios to some extent. As equivalized income is identical for all household members, we cannot control for spouse's income lnw_i^P in Eqs. (6)–(8). Because of this, Eq. (4) becomes identical to (6), and scenario (ii) is identical to (i).

Focusing on the remaining results for equivalized household income in comparison to individual net income in Fig. 5, we find a lower IOR in all specifications for household equivalized income. This is not too surprising as equalizing the income shades potential income differences between spouses, such that the baseline case hovers around 20 percent. However, we do find additional explanatory power for spouses' variables in the remaining scenarios. This effect is even larger compared to individual income (increase of up to 20 percentage points). Moreover, increases in the IOR are stronger over time in the case of no responsibility compared to the baseline case with equivalized income as well as compared to individual income.

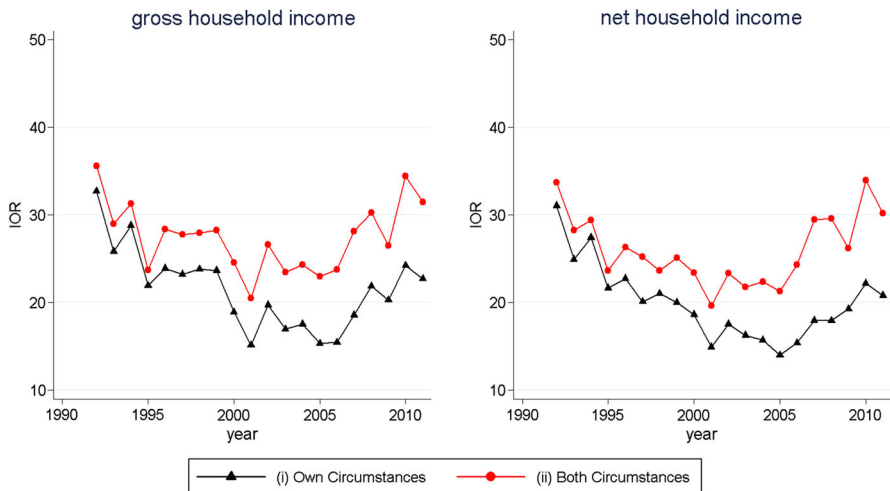


Fig. 6 IOR in GE(0) for gross vs. net household income

Bringing the analysis to the household level, one can also look at joint household income in gross and net terms. We include the circumstances of the household head in the baseline case and add those of the partner as a second scenario. It is a well known fact that inequality between households is lower compared to inequality between individuals, as intra-household inequality accounts for a substantial amount of overall inequality. Therefore, it is less of a surprise that IOP is also lower in the household context. However, it is noteworthy that IOP has a U-shaped pattern and the additional explanatory power of the spouse's circumstances increases over time (Fig. 6).

In the baseline results, calculations were carried out using the MLD as the measure of inequality. This, however, is only one special case from the generalized entropy indices with parameter zero (GE(0)). Pistolessi (2009) for example uses a broad range of inequality measures, including the Gini coefficient. Therefore, we run the analysis using different GE measures with different sensitivity for different parts of the income distribution [GE(-1), G(1), G(2)] as well as the Gini coefficient as inequality measures¹⁴ in order to compare the results both for gross and net income. Generally, the results point in the same direction. In particular, the increased explanatory power of partner variables holds for all inequality measures. Interestingly, overall results for GE(1), GE(2), and the Gini coefficient are higher compared to the MLD (GE(0)), while results for GE(-1) are lower. That is, the IOR is lower when more emphasis is put on the bottom rather than the top of the distribution (Fig. 7).

¹⁴ Note that these measures do not fulfill the axiom of path-independent decomposability and hence the results should be interpreted with caution.

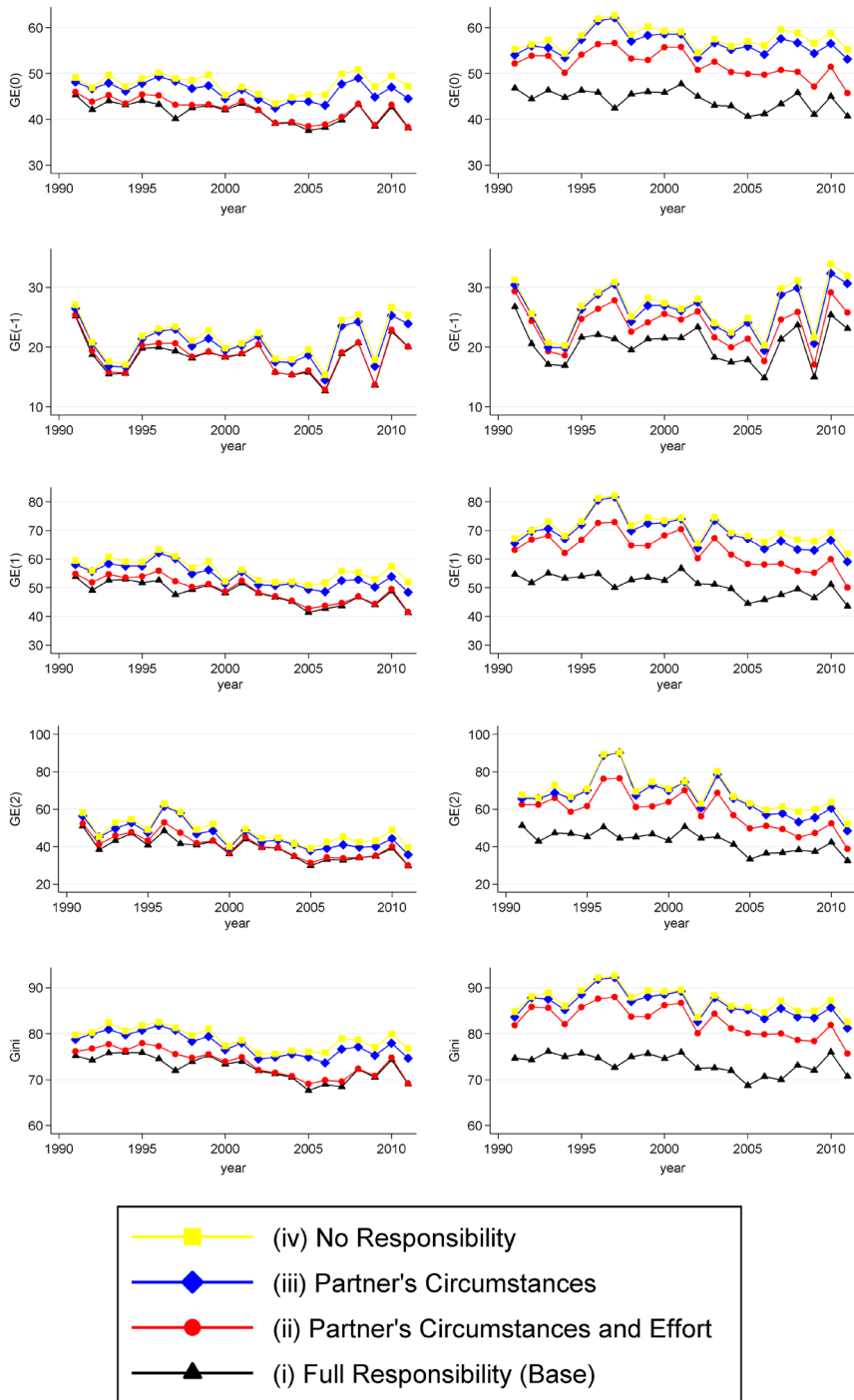


Fig. 7 IOR for additional measures—gross vs. net

Table 1 Correlations of selected variables

Year	Annual earnings	Hourly wages	Education
1992	−0.396	0.080	0.561
1994	−0.368	0.131	0.605
1996	−0.475	0.061	0.572
1998	−0.395	0.119	0.567
2000	−0.391	0.032	0.569
2002	−0.355	0.140	0.563
2004	−0.322	0.070	0.577
2006	−0.338	0.067	0.549
2008	−0.277	0.072	0.593
2010	−0.283	0.076	0.558
2011	−0.225	0.168	0.631

4.5 Assortative mating

In order to gain initial insight into the extent of assortative mating, we compare correlations of key variables displayed in Table 1. We find a negative correlation between spouses' annual earnings. Considering the differences in part-time work between female and male employees, this is not surprising. The negative correlation decreases over time which might stem from a decline of the male breadwinner model as more women start to work. Looking at hourly wages rather than annual earnings accounts for the difference in labor supply. Here, we find a small but persistent positive correlation in wages. According to the literature ([Schwartz and Mare 2005](#)), the main variable to measure assortative mating is education. Considering years of education, a positive correlation is found in all years with a range between 0.549 and 0.631. Hence, spouses tend to have similar years of education.

In order to investigate the role of assortative mating for IOp in couples, we follow the literature on assortative mating ([Burtless 1999](#); [Aaberge et al. 2005](#); [Pestel 2016](#)) and re-compute the IOp measures for the 4 different scenarios by randomly re-matching couples. The effect of assortative mating on inequality is then assessed by comparing the obtained IOp measures for the re-matched sample to the results for the original sample.¹⁵ The resulting IOp measures are displayed in Fig. 8. Note that overall inequality does not change as we do not drop any individuals and the baseline case (without spousal variables) is also not affected. Since, by construction, the correlation between spouses' earnings in the random sample is (close to) zero, the differences between the scenarios disappear. Hence, this exercise shows that assortative mating is key for the conclusion that including a spouse's information in the analysis matters for IOp.

¹⁵ Note that we abstract from potential behavioral responses (such as labor supply) when facing a new partner with different characteristics ([Pestel 2016](#)). Furthermore, we only compare IOp measures in gross earnings as we would have to re-calculate the total tax burden of the new randomly matched couples in order to also analyze IOp in net earnings.

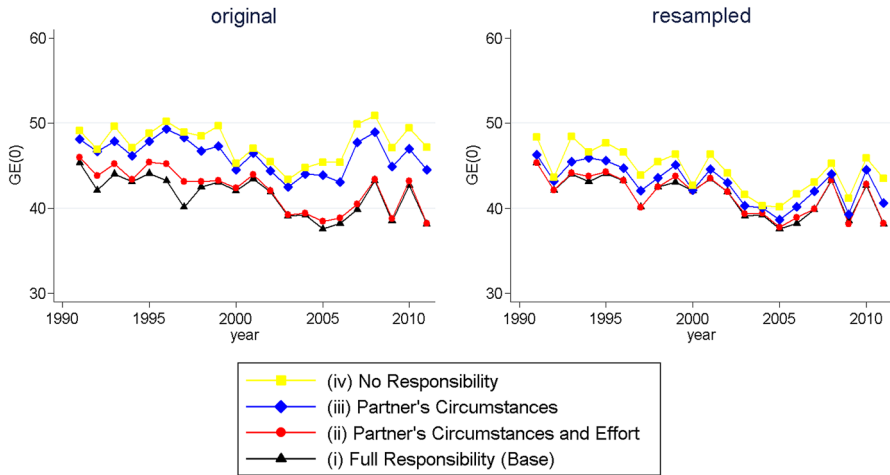


Fig. 8 IOR for gross annual income—rematched couples

5 Conclusion

Thus far, existing literature on IOp has failed to analyze the influence of the partner in measuring IOp of individuals in couples. Using German micro data from 1991 to 2011, we add to the literature by analyzing the effect of a spouse's circumstance, effort and earnings variables on individual earnings and hence IOp. As additional explanatory factors tend to increase lower bounds for IOp, we find that the spouse's variables do in fact have a significant effect on IOp measures. As the effect of spouse's income is declining over time, we suspect that assortative mating is playing a role in this context. The positive correlation between the education level of spouses and a diminishing negative correlation of earnings further emphasize this. Sensitivity analyses using household income concepts as well as further inequality measures apart from MLD suggest that our results are valid not only for the main specification of our analysis but rather hold for a more general case.

To summarize, we find that accounting for personal information of the spouse matters for IOp as our estimate of the IOR increases by up to 20 (35) percent for gross (net) earnings. The less the responsibility assumed for the spouse's variables, the higher the IOp measure. The question as to which scenario is most suitable cannot be answered without moral judgment and consideration of the context of the research question. Given that spousal variables were completely omitted from analysis in previous studies, the implicit assumption was full responsibility for all spousal variables. It might be questioned whether this omission was intentional or rather inadvertent. Taking into consideration the relationship between spousal earnings via joint income tax filing and different models for decision-making within the household and against the background of increased assortative mating and its potential role in changing inequality, explicitly accounting for the partner (or not) seems to be important when measuring IOp.

Appendix

A.1 Variables

See Table 2.

Table 2 Variable glossary

Variable	Description
Income	
Individual earnings	Gross and net annual individual earnings
HH equivalized income	OECD equivalence scale
Circumstance variables	
Gender	Male, female
Foreign origin	Nationality: native, foreign
East-Germany	Born in East- or West-Germany
Fathers occupation	Classified: blue collar, farmer, white collar, free-lancer and high qualified, civil servant
Fathers education	Classified: no degree, secondary, intermediate, upper secondary
Urbanization	Classified: countryside, small city, large city
Year of birth	In years
Height	Body height (in centimeter)
Effort variables	
Industry	Classified: public administration and social security; fishery, agriculture, forestry; energy, chemicals, steel; engineering; manufacturing; construction; wholesale, trade; transport; financial service; education; health service
Education	In years
Experience	In years, normalized at the age level
Working hours	In weekly hours

A.2 Descriptive statistics

See Tables 3 and 4.

Table 3 Descriptive statistics for basic variables

Variable	Mean	Sd	Min	Max
Income				
Gross income	30,547.48	26,270.32	25	900,562.9
Net income	21,832.4	17,053.6	22.15	505,810.9
Equivalized income	22,021.5	12,908.36	2828.89	373,132.7
Circumstance variables				
Female	0.5	0.5	0	1
Foreign	0.04	0.21	0	1
East-Germany	0.31	0.46	0	1
No degree	0.01	0.11	0	1
Secondary	0.76	0.43	0	1
Intermediate	0.12	0.33	0	1
College	0.11	0.31	0	1
Blue-collar	0.55	0.5	0	1
Farmer	0.04	0.19	0	1
White-collar	0.14	0.35	0	1
Professional	0.13	0.33	0	1
Self-employed	0.07	0.25	0	1
Civil servant	0.08	0.27	0	1
Countryside	0.41	0.49	0	1
City	0.38	0.49	0	1
Large city	0.21	0.41	0	1
Year of birth	1957	7.67	1929	1983
Effort variables				
Body height	173.22	9.08	139	205
Work experience	−0.06	0.93	−3.3	2.71
Weekly working hours	32.47	13.17	0.88	93.84
Education	12.7	2.66	7	18
Civil servant	0.1	0.3	0	1
Energy and mining	0.13	0.34	0	1
Engineering	0.07	0.26	0	1
Manufacturing	0.05	0.22	0	1
Construction	0.08	0.27	0	1
Sales	0.15	0.35	0	1
Transport	0.05	0.21	0	1
Financial	0.03	0.18	0	1
Service	0.13	0.34	0	1
Education	0.1	0.29	0	1
Health	0.11	0.31	0	1

Table 4 Descriptive statistics for IOP measures

Year	EO	IOR gross income				IOR net income			
		Case 1	Case 2	Case 3	Case 4	Case 1	Case 2	Case 3	Case 4
1991	0.29	45.35	45.96	48.14	49.13	46.82	52.19	54.01	55.28
1992	0.33	42.11	43.84	46.65	46.93	44.42	53.87	56.03	56.31
1993	0.33	44.03	45.25	47.90	49.64	46.35	53.88	55.62	57.31
1994	0.34	43.14	43.42	46.19	47.15	44.77	50.15	53.48	54.33
1995	0.33	44.09	45.44	47.87	48.85	46.28	54.06	57.41	58.30
1996	0.32	43.25	45.19	49.36	50.20	45.85	56.43	61.45	61.96
1997	0.32	40.13	43.14	48.32	48.93	42.37	56.67	62.12	62.66
1998	0.35	42.50	43.12	46.75	48.49	45.51	53.26	57.03	58.50
1999	0.33	43.07	43.23	47.33	49.73	45.99	52.90	58.32	60.26
2000	0.36	42.04	42.39	44.51	45.25	45.84	55.75	58.62	59.26
2001	0.35	43.46	43.95	46.49	47.08	47.69	55.82	58.56	58.99
2002	0.35	41.95	42.03	44.38	45.49	45.01	50.72	53.43	54.47
2003	0.35	39.09	39.23	42.50	43.40	43.04	52.55	56.70	57.51
2004	0.35	39.23	39.42	44.02	44.80	42.91	50.30	55.23	55.91
2005	0.37	37.55	38.47	43.89	45.42	40.63	49.87	55.90	56.95
2006	0.35	38.21	38.86	43.05	45.40	41.18	49.72	54.13	56.09
2007	0.39	39.86	40.50	47.72	49.88	43.37	50.77	57.62	59.56
2008	0.37	43.27	43.36	48.97	50.88	45.77	50.35	56.67	58.82
2009	0.37	38.49	38.77	44.87	47.12	41.03	47.10	54.37	56.58
2010	0.37	42.69	43.18	46.99	49.47	45.00	51.41	56.52	58.81
2011	0.37	38.12	38.22	44.52	47.19	40.71	45.70	53.14	55.20

A.3 Inequality of opportunity levels

The lower bound level of inequality of opportunity (IOL) is estimated by computing the MLD of the predicted values from the earnings equation. We do this for specifications Eqs. (4) and (6)–(8) using gross and net annual earnings. In addition, we distinguish between male and female individuals. The results for gross and net earnings are shown in Fig. 9. The black line displays the baseline case from Eq. (4). The red line corresponds to the model including spouse's circumstance variables shown by Eq. (6). The case of the model from Eq. (7), including both the spouse's circumstance and effort variables, is displayed by the blue line. Finally, the full specification including circumstances, effort and income variables of the partner corresponds to the yellow graph (Eq. (8)).

For the baseline case, the inequality of opportunity level is lower for net earnings than for gross earnings. Aside from this, the figures for IOL in net earnings are slightly higher. Regardless of the specification, IOL shows a slightly increasing trend, with a jump between 2006 and 2007. In terms of gross income, controlling for the spouse's income (case of responsibility for spouse's circumstances and effort, displayed by the

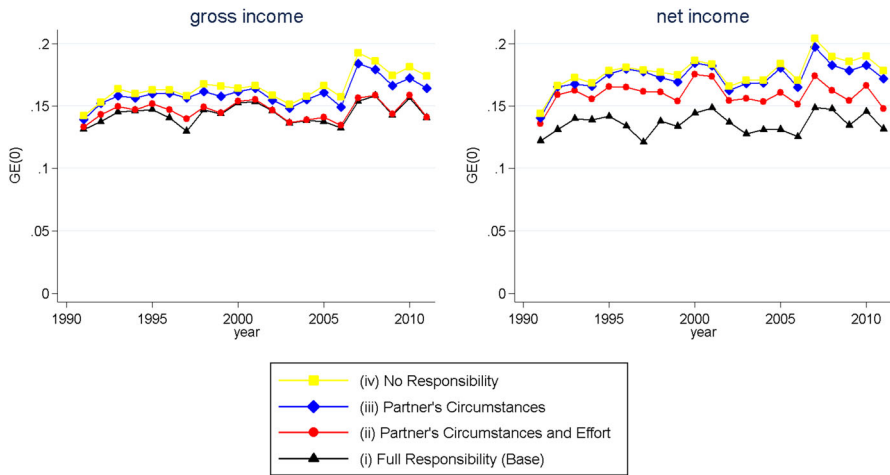


Fig. 9 IOL for annual income—gross vs. net

red graph) does not increase IOL. Using net earnings instead, we find a significant increase in explanatory power and hence also in IOL when controlling for the income of the partner. This indicates that it is an effect of the German tax system, as is already visible from the earnings regression. The effect, however, declines over time, visualized on the graph by the narrowing gap between the red and black lines. This may be due to the changing relation of spouses' incomes. We indeed find a negative, but in absolute terms decreasing, correlation between partners' incomes.¹⁶

The case of responsibility for spouse's circumstances represented by the blue graph shows mostly identical results for gross and net earnings.¹⁷ Both graphs show an increased IOL compared to the previous cases. Finally, the case of no responsibility, displayed by the yellow graphs, shows a similar development compared to the previous case. However, since 2005, the circumstance variables of the partner increase in explanatory power.

In order to further disentangle the development of IOL, we also consider male and female individuals separately. The results are shown in Fig. 10 for gross and net earnings as well as for male and female sub-samples. In general, we find that the personal information of the partner has more explanatory power for women than for men, thereby increasing our IOL measures in the different scenarios. Male IOL in gross and net earnings is fairly constant until 2008, with a sharp increase thereafter. For the female sub-sample, we find greater fluctuations over time and also higher IOL for all specifications except the baseline case. In contrast to the development of male IOL, the inequality of opportunity level for women decreases after 2008. When comparing male

¹⁶ The literature on assortative mating shows an increase in assortative mating in education over time that also reflects in earnings (Schwartz and Mare 2005). A detailed discussion on the importance of assortative mating is conducted in 4.5.

¹⁷ The case of responsibility for spouse's circumstances is implemented by controlling for spouse's effort and income variables in Eq. (7).

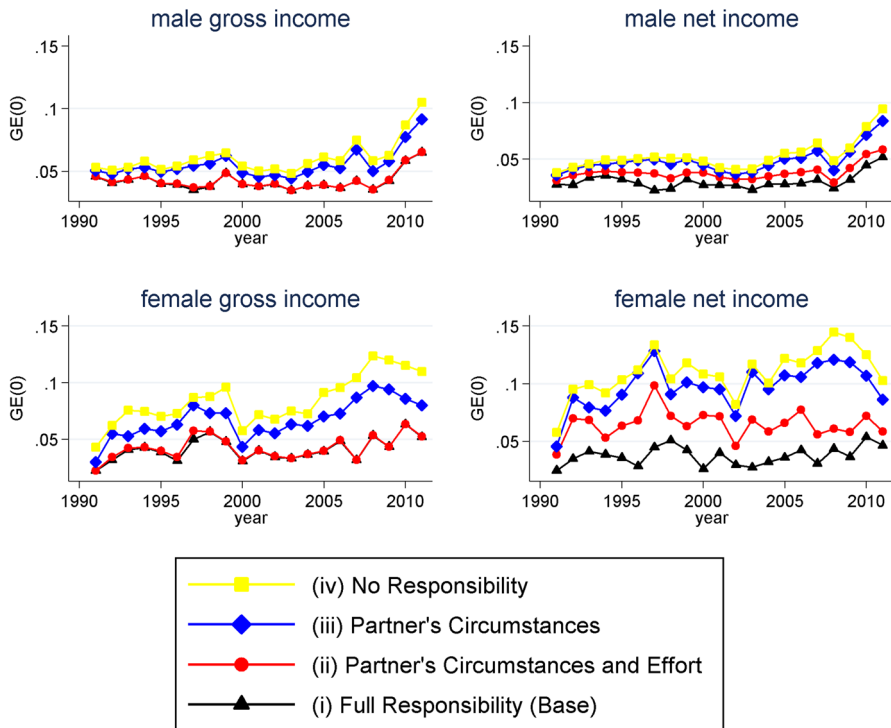


Fig. 10 IOL for annual income—male vs. female net

and female sub-samples in terms of net earnings, the case of responsibility for spouse's circumstance (red graph) and effort variables yields interesting results. Controlling for spouse's earnings only slightly increases IOL for men, while there is a significantly larger effect for women. However, following a peak in 1997, the divergence between the black and red graph diminishes over time. This again indicates an effect of the income tax splitting as well as changing correlation in spouses' earnings.

References

- Aaberge R, Aslaksen I, Wennemo T (2005) 'Birds of a feather flock together': The impact of choice of spouse on family labor income inequality. *Labour* 19(3):491–515
- Aaberge R, Mogstad M, Peragine V (2011) Measuring long-term inequality of opportunity. *J Public Econ* 95(3):193–204
- Arneson RJ (1989) Equality and equal opportunity for welfare. *Philos Stud* 56(1):77–93
- Blackburn ML (2007) Estimating wage differentials without logarithms. *Labour Econ* 14(1):73–98
- Bourguignon F, Ferreira FH, Menendez M (2007) Inequality of opportunity in brazil. *Rev Income Wealth* 53(4):585–618
- Browning M, Chiappori P-A, Weiss Y (2013) *Family economics*. Cambridge University Press, Cambridge
- Burtless G (1999) Effects of growing wage disparities and changing family composition on the us income distribution. *Eur Econ Rev* 43(4):853–865
- Checchi D, Peragine V (2010) Inequality of opportunity in Italy. *J Econ Inequal* 8(4):429–450

- Chiappori PA, Meghir C (2015) Intrahousehold inequality. In: Handbook of income distribution, vol. 2B. North Holland, pp 1369–1418
- Cohen GA (1989) On the Currency of Egalitarian Justice. *Ethics* 99(4):906–944
- Dworkin R (1981a) What is Equality? Part 1: Equality of Welfare. *Philos Public Aff* 10(3):185–246
- Dworkin R (1981b) What is Equality? Part 2: Equality of Resources. *Philos Public Aff* 10(4):283–345
- Ferreira FHG, Peragine V (2015) Equality of opportunity: Theory and evidence. In: World Bank Policy Research Paper (7217)
- Ferreira FH, Gignoux J (2011) The measurement of inequality of opportunity: theory and an application to Latin America. *Rev Income Wealth* 57(4):622–657
- Fleurbaey M (1995) Three solutions for the compensation problem. *J Econ Theory* 65(2):505–521
- Fleurbaey M, Maniquet F (2008) Fair social orderings. *Econ Theor* 34(1):25–45
- Fleurbaey M, Peragine V (2013) Ex ante versus ex post equality of opportunity. *Economica* 80(317):118–130
- Foster JE, Shneyerov AA (2000) Path independent inequality measures. *J Econ Theory* 91(2):199–222
- Haisken De-New J., Frick J (2005) Dtc-desktop compendium to the german socio-economic panel study (gsoep). Deutsches Institut für Wirtschaftsforschung, Berlin
- Lefranc A, Pistolesi N, Trannoy A (2008) Inequality of opportunities vs. inequality of outcomes: are western societies all alike? *Rev Income Wealth* 54(4):513–546
- Lefranc A, Pistolesi N, Trannoy A (2009) Equality of opportunity and luck: Definitions and testable conditions, with an application to income in france. *J Public Econ* 93(11):1189–1207
- Lise J, Seitz S (2011) Consumption inequality and intra-household allocations. *Rev Econ Stud* 78(1):328–355
- Niehues J, Peichl A (2014) Upper bounds of inequality of opportunity: theory and evidence for germany and the us. *Soc Choice Welf* 43(1):73–99
- Nozick R (1974) *Anarchy, State, and Utopia*. Basic Books, New York
- Pestel N (2016) Beyond inequality accounting: marital sorting and couple labor supply. *Economica* (forthcoming)
- Pistolesi N (2009) Inequality of opportunity in the land of opportunities, 1968–2001. *J Econ Inequal* 7(4):411–433
- Ramos X, Van de gaer D (2016) Empirical approaches to inequality of opportunity: principles, measures, and evidence. *J Econ Surv* (forthcoming)
- Roemer J (1998) *Equal Oppor*. Harvard University Press, Cambridge
- Roemer JE (1993) A pragmatic theory of responsibility for the egalitarian planner. *Philosophy & Public Affairs*, USA
- Roemer J, Trannoy A (2016) Equality of opportunity: theory and measurement. *J Econ Lit* (forthcoming)
- Schwartz CR, Mare RD (2005) Trends in educational assortative marriage from 1940 to 2003. *Demography* 42(4):621–646
- Shorrocks AF (1980) The class of additively decomposable inequality measures. *Econometrica*:613–625
- Socio-Economic Panel (SOEP) (2013) Data for years 1984–2012. Version 29, SOEP. doi:[10.5684/soep.v29](https://doi.org/10.5684/soep.v29)
- Van de gaer D (1993) Equality of opportunity and investment in human capital. Dissertation. KU Leuven
- van Parijs P (1995) Social justice and individual ethics*. *Ratio Juris* 8(1):40–63
- Wagner GG, Frick JR, Schupp J (2007) The german socio-economic panel study (soep)-evolution, scope and enhancements. *Schmollers Jahrb*. 127:139–170

